

When the data cannot travel, send the analysis instead

Global standards for federated health data science: the GA4GH approach



TEAM 3: ROSETTA · Interoperability · HDR UK Black Internship Programme 2026

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1 YOUR FLATMATE

She is rushed to A&E with severe stomach pain and is not responding to any antibiotic she has been given. To find out what is growing in her gut, a sample is sequenced.

That sequence has to be compared against a far larger reference collection, held by a research group in another country. **It cannot come to her. It is too large to move, governed by different data-sharing laws, and stored in a different format.**

Building a one-off system to bridge the two could take months. She does not have months.

2 WHY THIS KEEPS HAPPENING

THE CHARGING PORT

How many people bought a phone because of the charging port? Exactly. Rival technology companies and the European Commission had the same thought, and they made USB-C the standard.

That is a pre-competitive standard.

HEALTH RESEARCH HAD NO EQUIVALENT

So every collaboration was built from scratch, which limits the progress of research and human health care.

The agreement is the thing that was missing, not the technology.

3 WHO AGREES THE STANDARDS

The Global Alliance for Genomics and Health (GA4GH) is an international community that develops shared standards, policy frameworks and tools for genomic and health data.

It holds no data itself. Hospitals and research groups keep their own, and remain responsible for it. GA4GH's job is to agree the common rules so nobody has to invent them twice.

4 THE STANDARD WE STUDIED

You want to bake a cake. But the ingredients are in Australia. So you send the recipe to Australia. The cake is baked there and is then sent to you in the UK.

GA4GH's **Task Execution Service (TES)** is the recipe: one agreed structure for describing a computational task, what data it needs, what software runs and what comes back, so any compute system can read it.

TES is only one standard, alongside WES (whole workflows), Beacon (finding data) and Passports (access).

5 WHO IS PUTTING IT TO WORK

ELIXIR

Europe's life-science research infrastructure, organised as national Nodes.

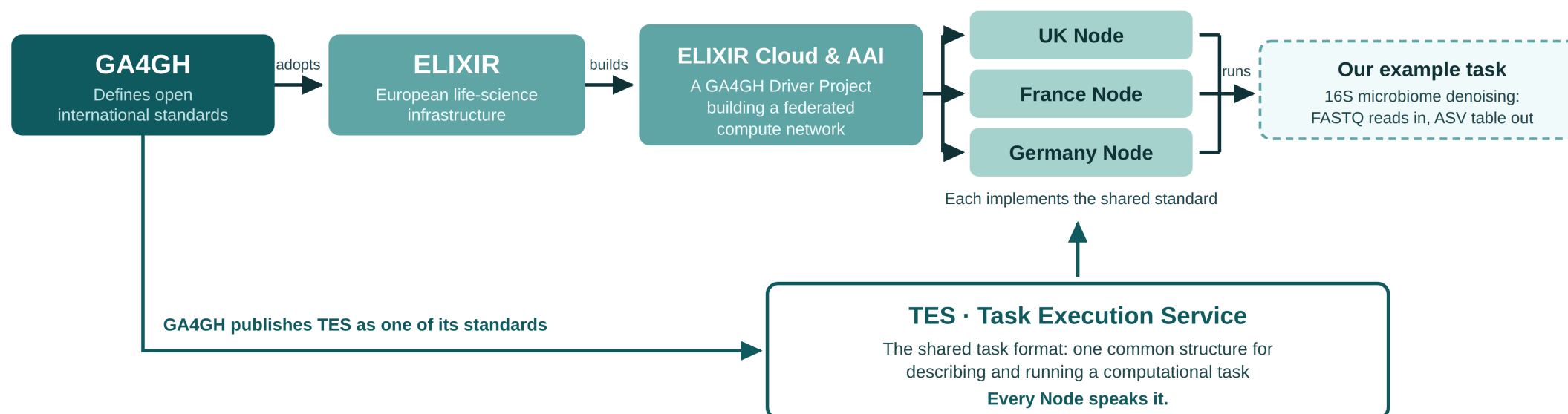
It has worked with GA4GH since **2013**, formalised as a Strategic Partnership in **2019**.

ELIXIR CLOUD & AAI

A GA4GH **Driver Project**, a real initiative that helps test and implement GA4GH standards.

It is **building** a federated compute network across those Nodes, with TES at its core.

6 HOW IT FITS TOGETHER



7 BACK TO YOUR FLATMATE

With a shared standard in place, nobody builds anything from scratch. Her lab describes the analysis *once*, as a TES task document.

It travels to the Node holding the reference collection, runs beside the data, and only the result comes back. *The data never moves.*

Illustrative scenario. Sequencing identifies which bacteria are present; treatment decisions rest on wider clinical testing.

8 WHAT OUR TEAM BUILT

BUILT AND VALIDATED

- An example TES task document, a three-step microbiome analysis, validated against the official GA4GH schema with no violations
- A plain-language explainer, a ten-question FAQ and a 35-term glossary
- A source log tying 25 claims to the official page each came from

AN IMPORTANT LIMIT

Validation proves the document's **shape** is right, not its **science**. A typo in a filename would pass every check and fail only when the task ran.

Our example uses synthetic data and has not been run on a live service.

The report

Three-minute elevator pitch

9 EVERY CLAIM TRACED

- The scenario was chosen by the team, not assigned: three candidates proposed by our medical doctor, all read, then voted on
- **25 factual claims** mapped to the exact official page each came from, with access dates
- Four claims listed **separately**, because no single page states them, including the TES versus WES contrast
- We name what we did not verify rather than implying that we did

10 THE REPORT CHECKS ITSELF

- The JSON is read from disk at knit time, so the appendix and the validation can never disagree
- Validation runs **inside the document**, rather than being reported from elsewhere
- A manifest states whether every required file was found
- Five files, one folder, knits from a fresh session

11 THE SAME PROBLEM, CLOSER TO HOME

DARE UK funded **TRE-FX** to demonstrate analysis across a federation of Trusted Research Environments, so that data scattered across the UK could answer urgent questions. That is the same problem shape TES addresses.

We did not verify which standards TRE-FX adopted, and make no claim that it uses GA4GH products.

Next: submit the document to a sandbox implementation such as Funnel or TESK; give a second standard the same treatment; test the explainer with a patient and public involvement group.

! ONE THING TO TAKE AWAY

Our task document passes every check the schema can make, and would still fail on a single mistyped filename.

A standard tells you the shape is right. It does not tell you the science is. That distinction is the most useful thing we learned.